Single categorical predictor: Interpreting the coefficients

MATH 213

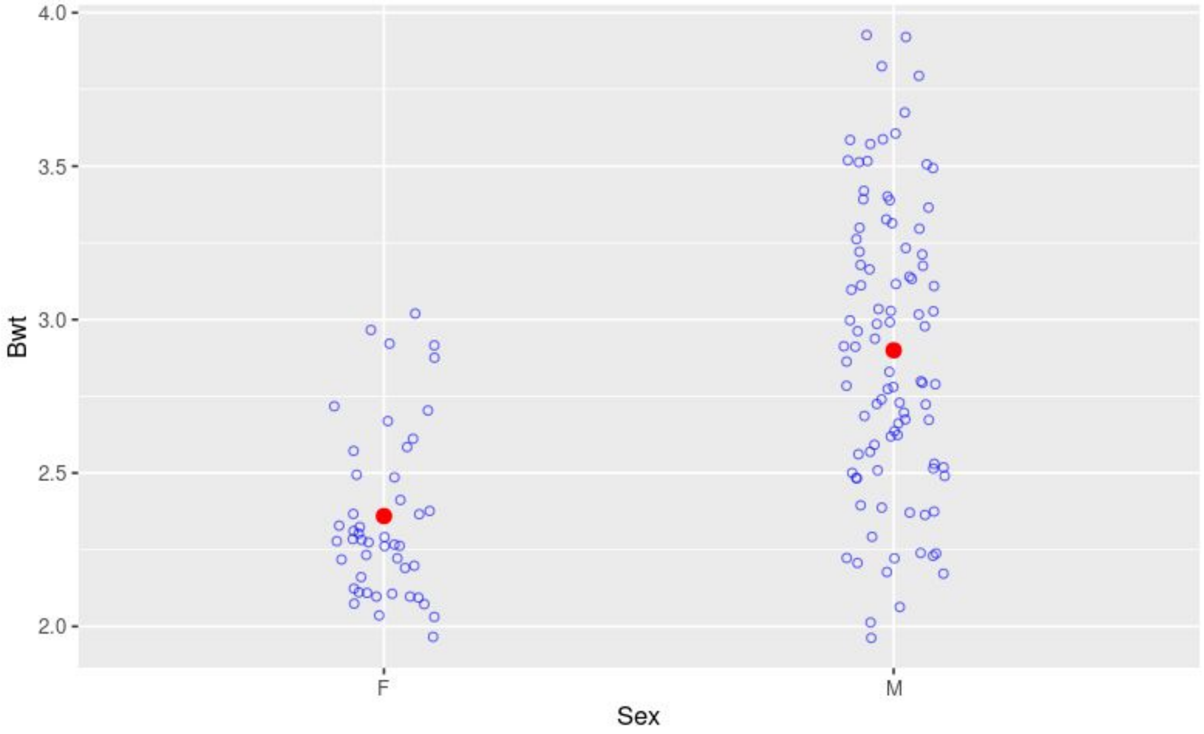
Group 1 - Question 1

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (`Hwt`), in grams, the body weight (`Bwt`) of each cat, in kg, and the sex of each cat, (`Sex`).

Here is a graph of the cats' body weights grouped by sex.

a. What am I using as a predictor variable?

a. What am I trying to predict? (What is the response variable?)



Here is some summary output for a regression model I made that matches the situation in the previous graph.

c. Write the linear regression equation.

d. Write a sentence of interpretation for the “intercept” coefficient in the model.

e. Write a sentence of interpretation for the “slope” coefficient in the model.

```{r}
cats_model2 <-lm(Bwt~Sex, data = cats)
summary(cats_model2)
```

```
Call:
lm(formula = Bwt ~ Sex, data = cats)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.90000 -0.25957 -0.05957  0.30000  1.00000 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.35957    0.06051  38.997  < 2e-16 ***
SexM         0.54043    0.07372   7.331 1.59e-11 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4148 on 142 degrees of freedom
Multiple R-squared:  0.2745,	Adjusted R-squared:  0.2694 
F-statistic: 53.74 on 1 and 142 DF,  p-value: 1.59e-11
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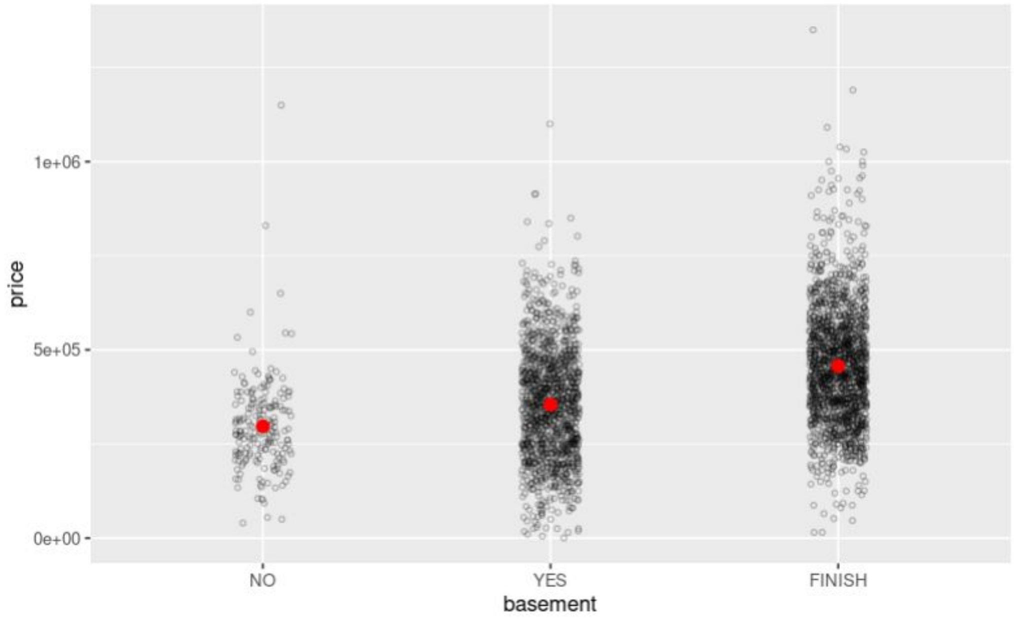
Question 2

I have a large dataset on house prices in Frederick from 2019. Two of the variables in the dataset are:

`price`: the sales price of the property

`basement`: whether the building has a finished basement (FINISH), unfinished basement (YES), or no basement (NO). **Read this again - it is sometimes misinterpreted!**

a. Does it look as though having a basement, finished or unfinished, makes a difference to the sales price of a house in Frederick?



Here is some summary output for a regression model I made.

Notice that when R made this model, it created two new variables: basementYES and basement FINISH. Here is what those mean:

basementYES takes value 1 if basement == YES and it takes value 0 otherwise.

basementFINISH takes value 1 if basement == FINISH and it takes value 0 otherwise.

c. What must the values of basementYES and basementFINISH be for a house that does not have a basement?

```
Call:
lm(formula = price ~ basement, data = frederick_2019)

Residuals:
    Min      1Q  Median      3Q     Max 
-441858 -104402   -6863   88156  893142 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)      296801      10619  27.950  < 2e-16 ***
basementYES       57601      11356   5.072 4.15e-07 ***
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Residual standard error: 148300 on 3281 degrees of freedom
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F-statistic: 236.8 on 2 and 3281 DF,  p-value: < 2.2e-16
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d. Write the regression equation for this model.

e. You should have *three* coefficients in your equation. Write a sentence of interpretation for each one.

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Here is a graph of the cats' body weights grouped by sex.

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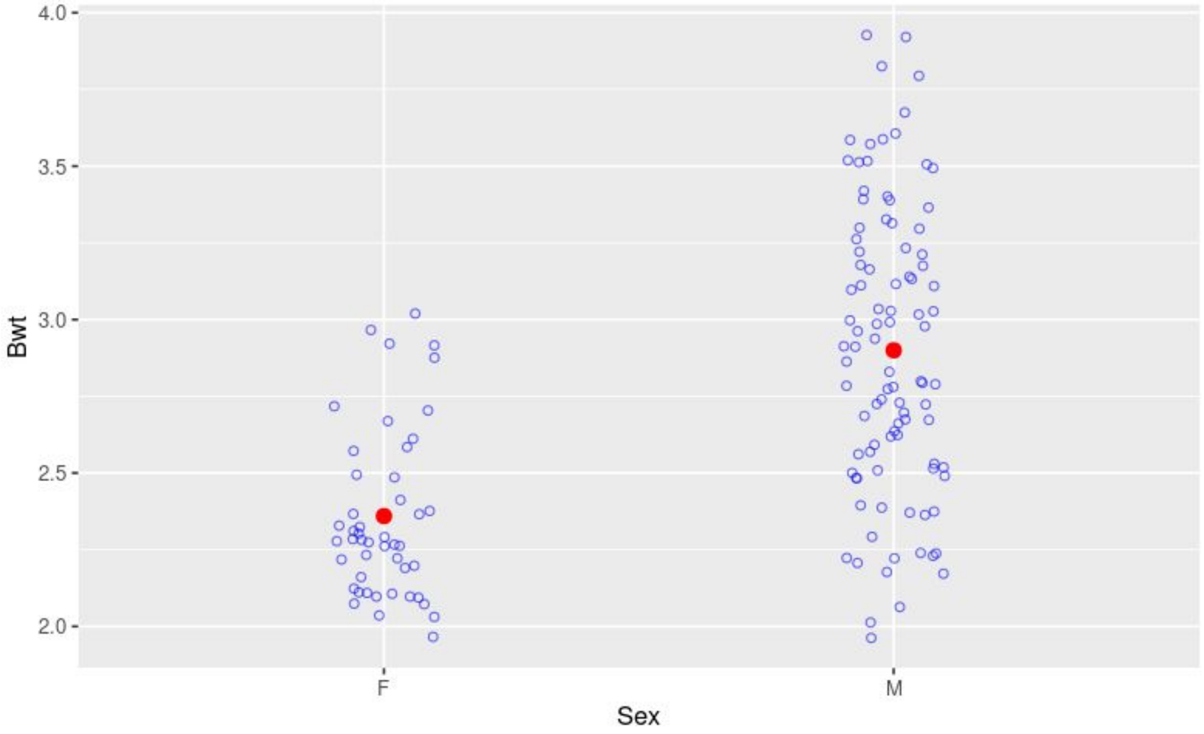

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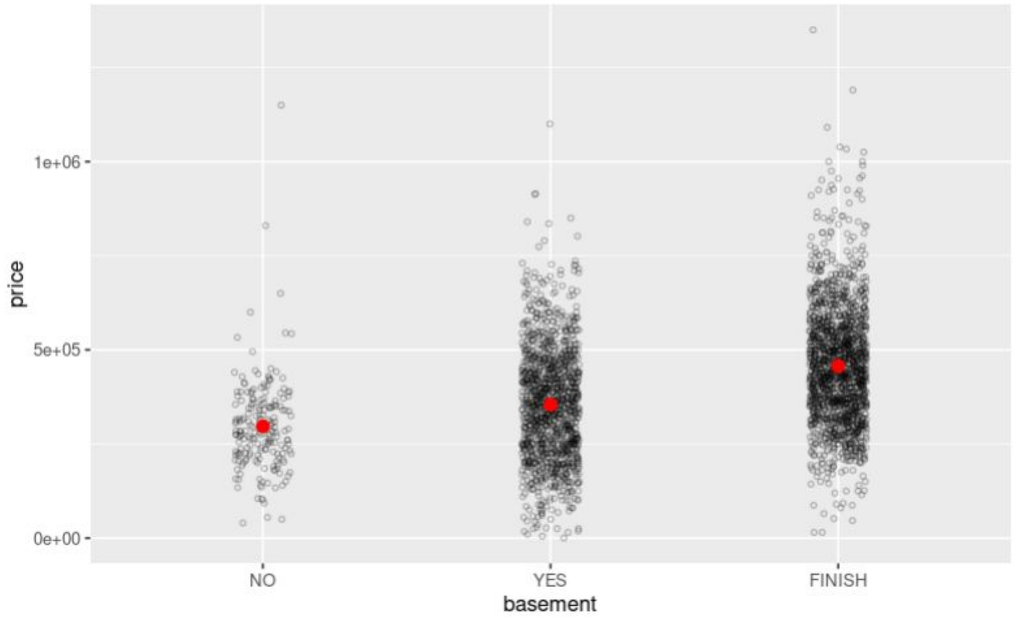

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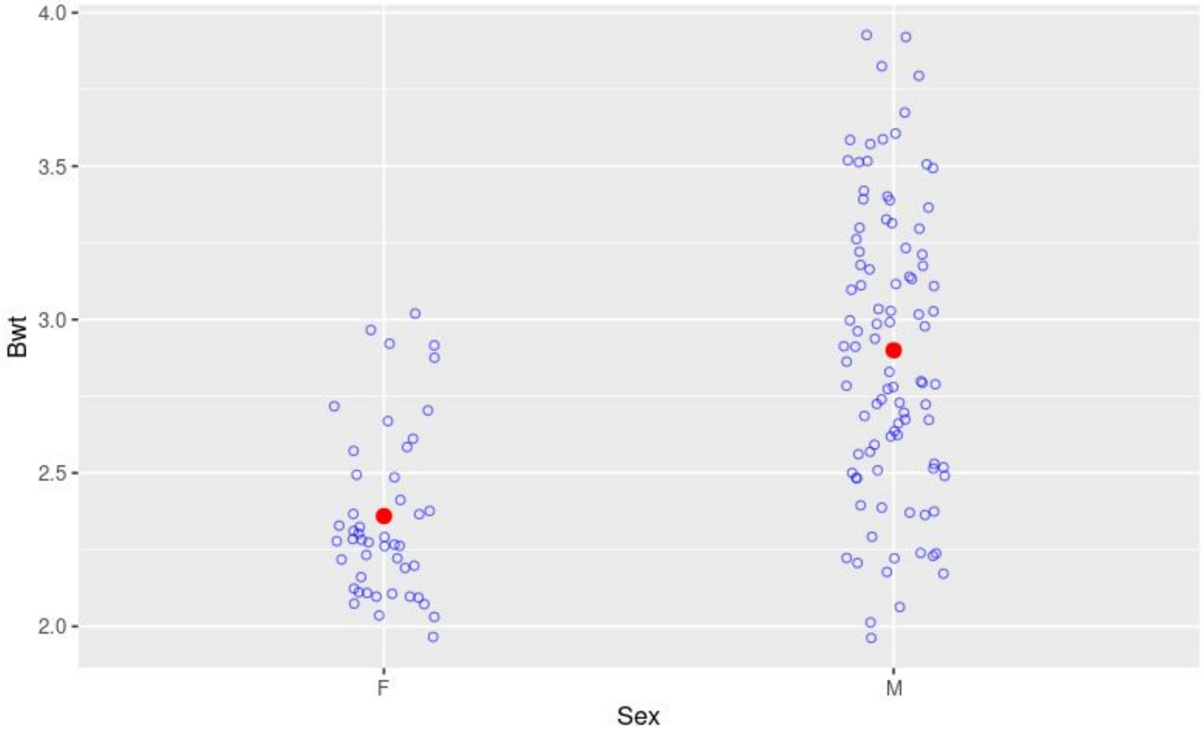
Here is a graph of the cats' body weights grouped by sex.



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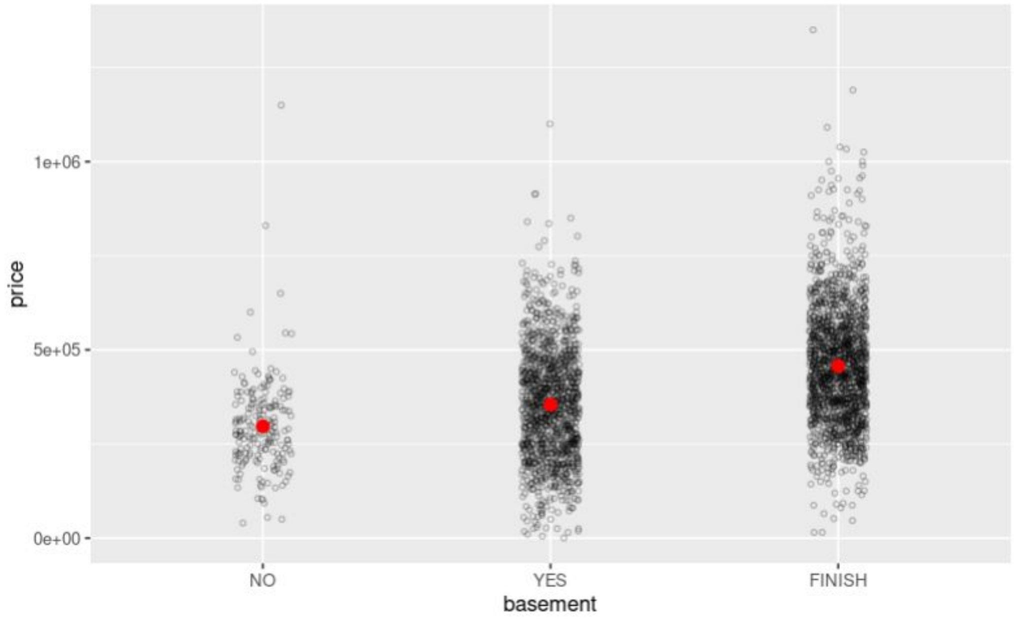

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Here is a graph of the cats' body weights grouped by sex.

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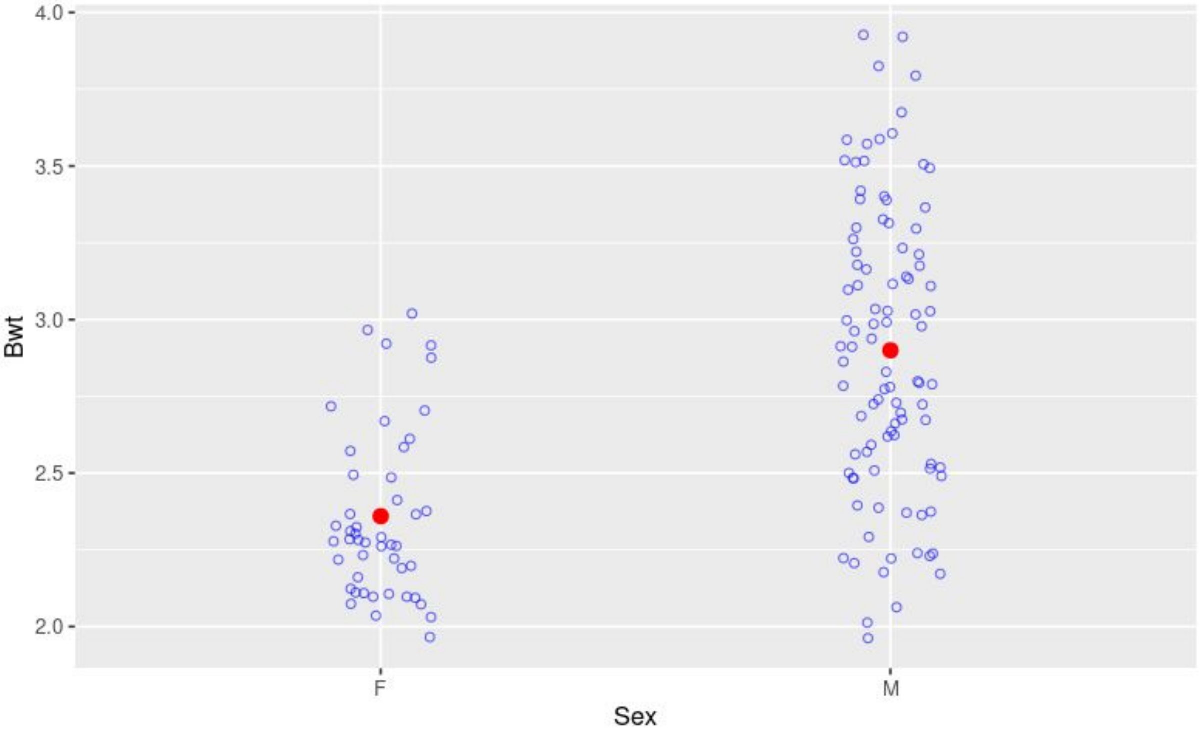

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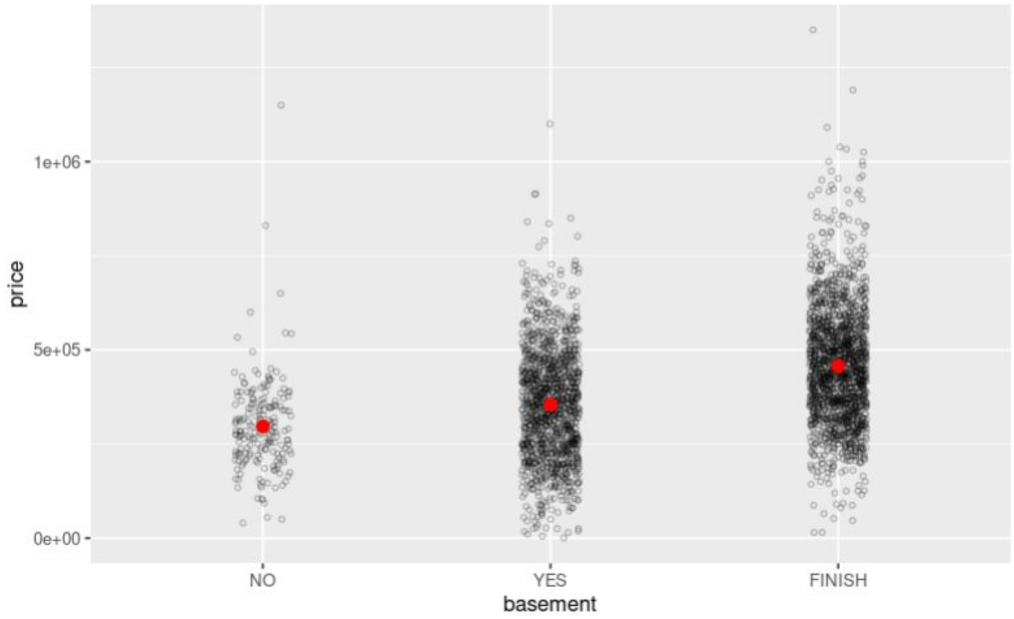

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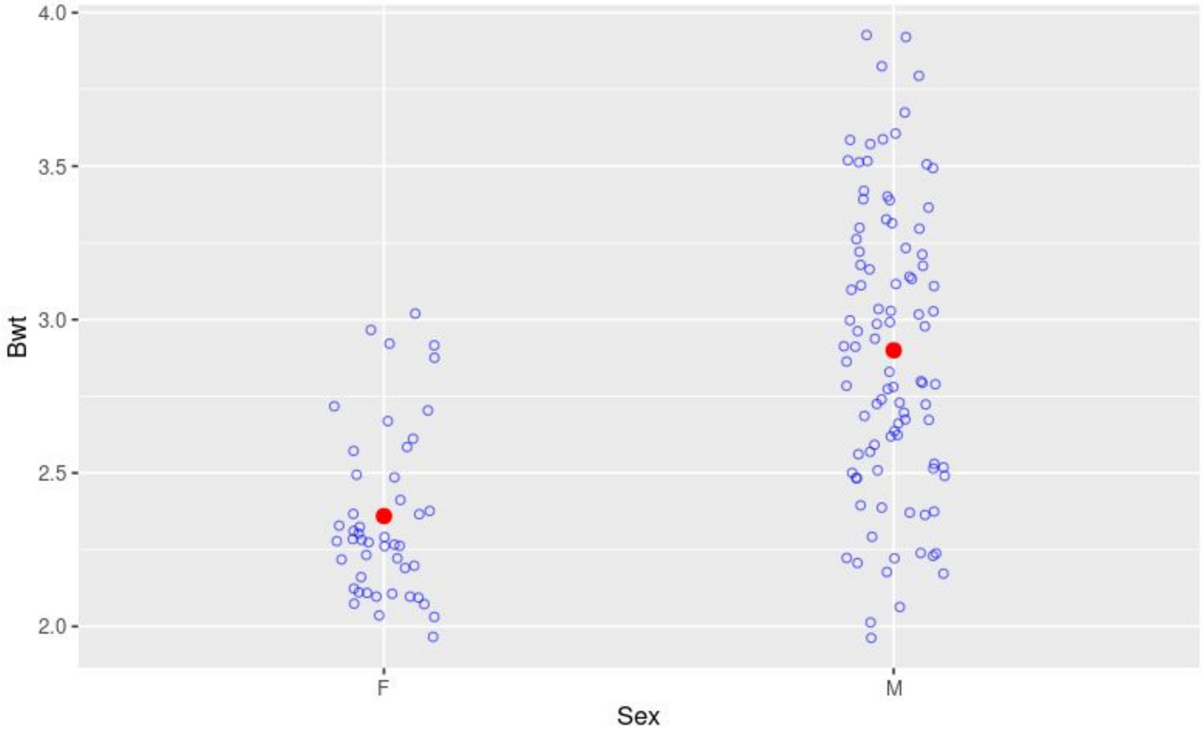
Here is a graph of the cats' body weights grouped by sex.



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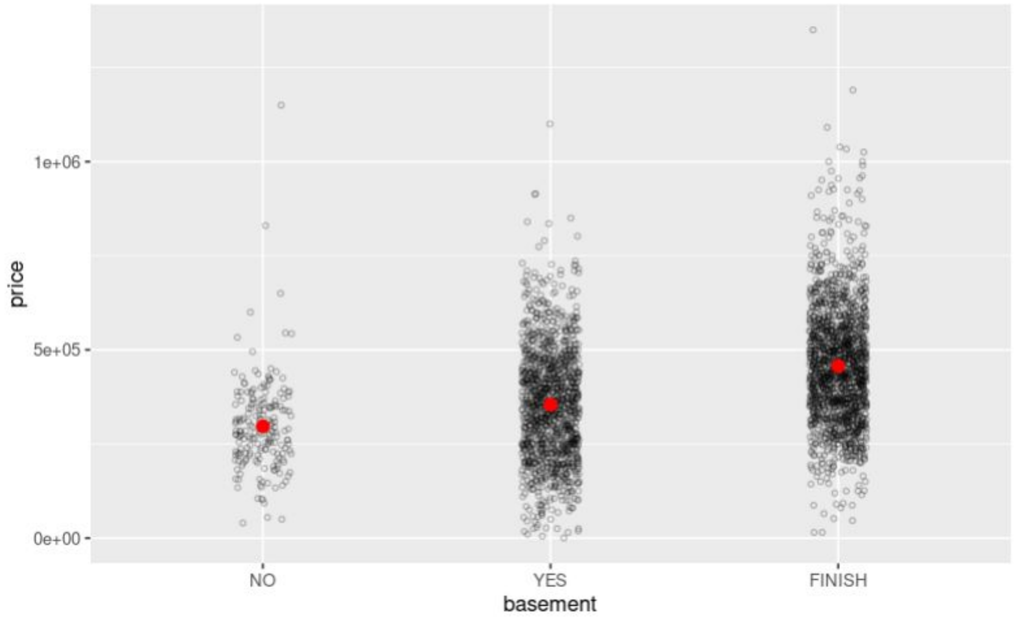

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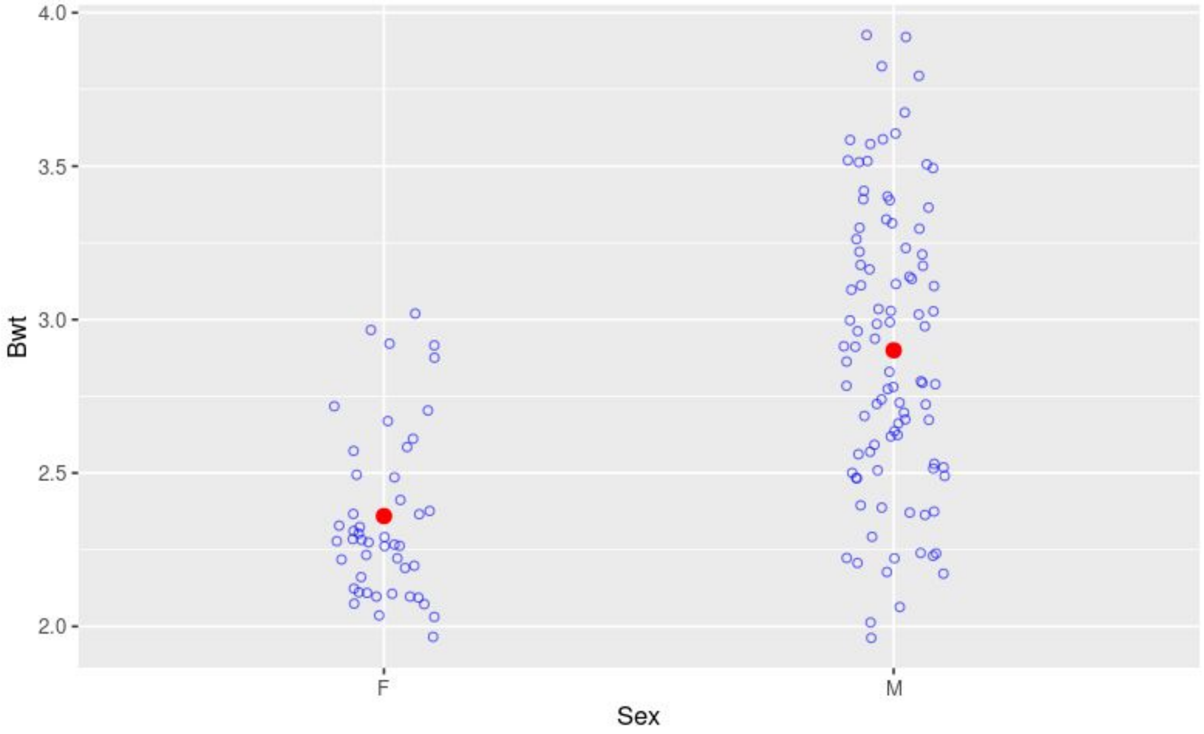
Here is a graph of the cats' body weights grouped by sex.



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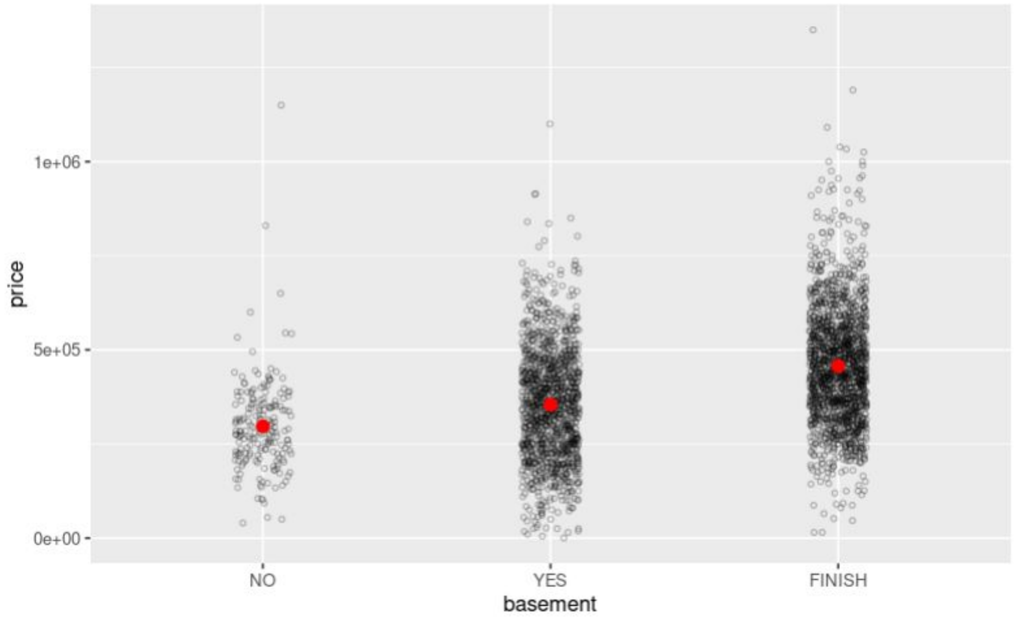

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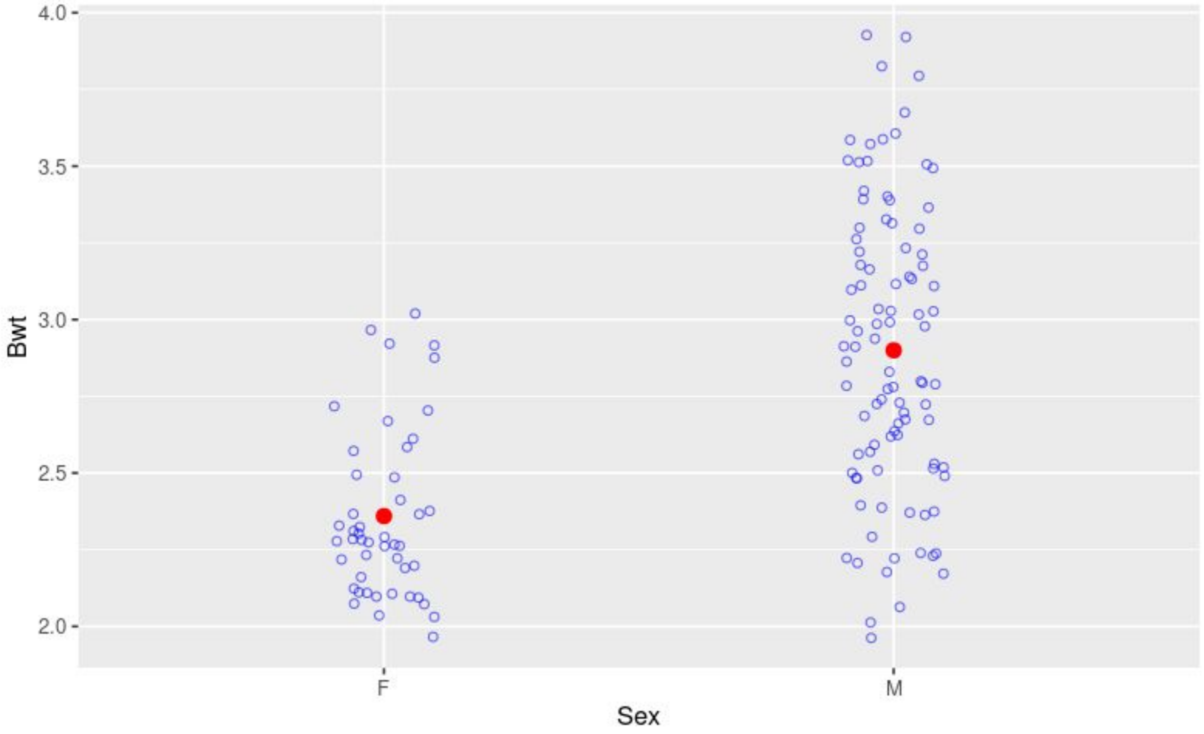

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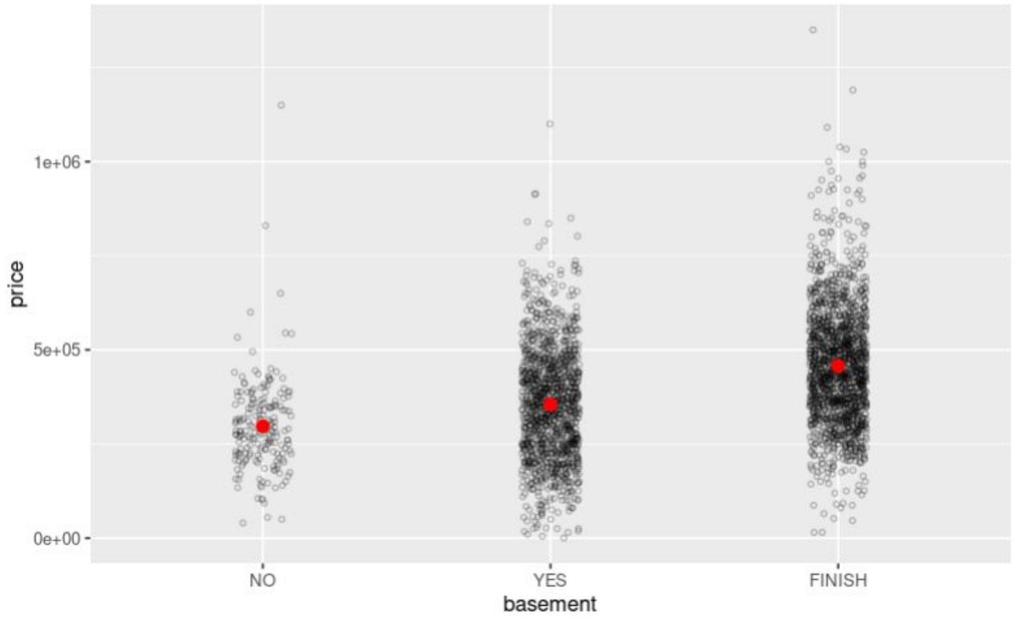


Here is some summary output for a regression model I made.

Notice that when R made this model, it created two new variables: basementYES and basement FINISH. Here is what those mean:

basementYES takes value 1 if basement == YES and it takes value 0 otherwise.

basementFINISH takes value 1 if basement == FINISH and it takes value 0 otherwise.

c. What must the values of basementYES and basementFINISH be for a house that does not have a basement?

```
Call:
lm(formula = price ~ basement, data = frederick_2019)

Residuals:
    Min      1Q  Median      3Q     Max 
-441858 -104402   -6863   88156  893142 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)      296801      10619  27.950  < 2e-16 ***
basementYES       57601      11356   5.072 4.15e-07 ***
basementFINISH   160057      11201  14.290  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 148300 on 3281 degrees of freedom
Multiple R-squared:  0.1261,	Adjusted R-squared:  0.1256 
F-statistic: 236.8 on 2 and 3281 DF,  p-value: < 2.2e-16
```

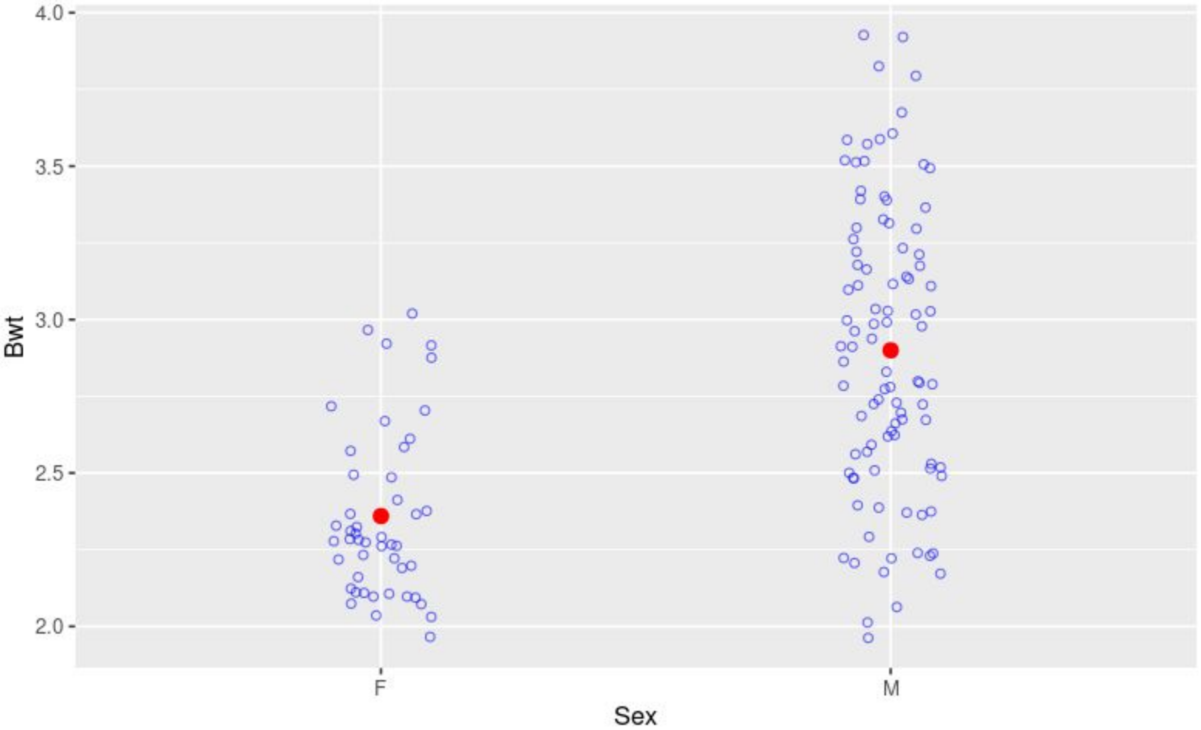
d. Write the regression equation for this model.

e. You should have *three* coefficients in your equation. Write a sentence of interpretation for each one.

Group 8 - Question 1

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (`Hwt`), in grams, the body weight (`Bwt`) of each cat, in kg, and the sex of each cat, (`Sex`).

Here is a graph of the cats' body weights grouped by sex.



a. What am I using as a predictor variable?

a. What am I trying to predict? (What is the response variable?)

Here is some summary output for a regression model I made that matches the situation in the previous graph.

c. Write the linear regression equation.

d. Write a sentence of interpretation for the “intercept” coefficient in the model.

e. Write a sentence of interpretation for the “slope” coefficient in the model.

````
```{r}
cats_model2 <-lm(Bwt~Sex, data = cats)
summary(cats_model2)
```
````

```
Call:
lm(formula = Bwt ~ Sex, data = cats)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.90000 -0.25957 -0.05957  0.30000  1.00000 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.35957    0.06051  38.997  < 2e-16 ***
SexM         0.54043    0.07372   7.331 1.59e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4148 on 142 degrees of freedom
Multiple R-squared:  0.2745,	Adjusted R-squared:  0.2694 
F-statistic: 53.74 on 1 and 142 DF,  p-value: 1.59e-11
```

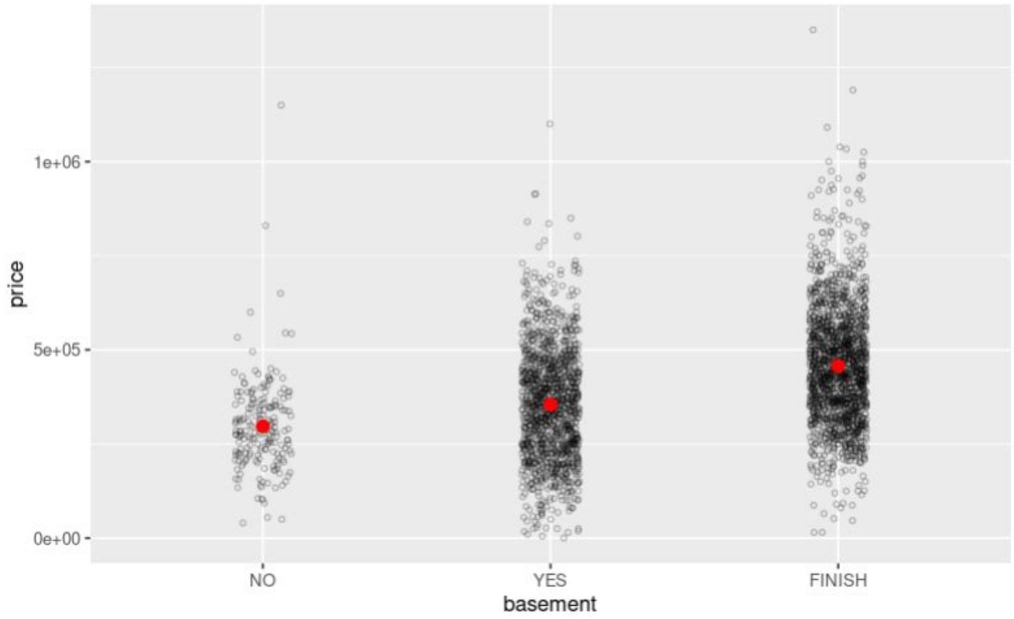
Question 2

I have a large dataset on house prices in Frederick from 2019. Two of the variables in the dataset are:

`price`: the sales price of the property

`basement`: whether the building has a finished basement (FINISH), unfinished basement (YES), or no basement (NO). **Read this again - it is sometimes misinterpreted!**

a. Does it look as though having a basement, finished or unfinished, makes a difference to the sales price of a house in Frederick?



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Notice that when R made this model, it created two new variables: basementYES and basement FINISH. Here is what those mean:

basementYES takes value 1 if basement == YES and it takes value 0 otherwise.

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c. What must the values of basementYES and basementFINISH be for a house that does not have a basement?

```
Call:
lm(formula = price ~ basement, data = frederick_2019)

Residuals:
    Min      1Q  Median      3Q     Max 
-441858 -104402   -6863   88156  893142 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)      296801      10619  27.950  < 2e-16 ***
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---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 148300 on 3281 degrees of freedom
Multiple R-squared:  0.1261,	Adjusted R-squared:  0.1256 
F-statistic: 236.8 on 2 and 3281 DF,  p-value: < 2.2e-16
```

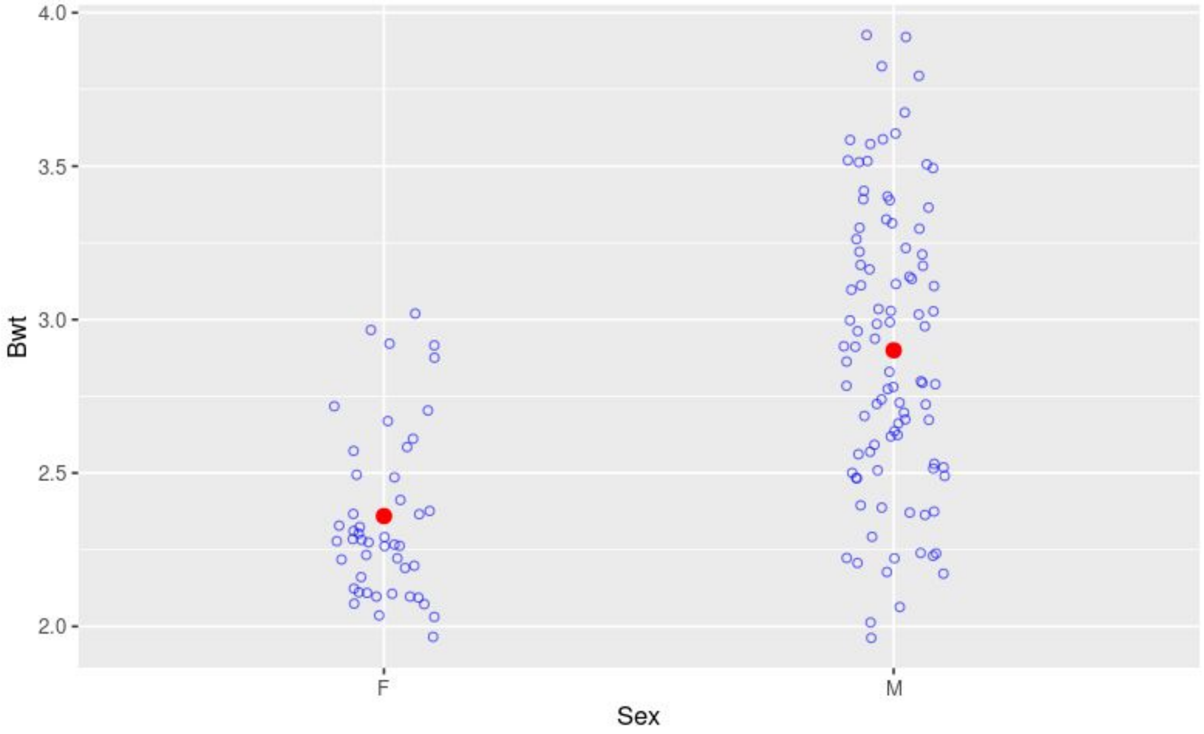
d. Write the regression equation for this model.

e. You should have *three* coefficients in your equation. Write a sentence of interpretation for each one.

Group 9 - Question 1

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (`Hwt`), in grams, the body weight (`Bwt`) of each cat, in kg, and the sex of each cat, (`Sex`).

Here is a graph of the cats' body weights grouped by sex.



a. What am I using as a predictor variable?

a. What am I trying to predict? (What is the response variable?)

Here is some summary output for a regression model I made that matches the situation in the previous graph.

c. Write the linear regression equation.

d. Write a sentence of interpretation for the “intercept” coefficient in the model.

e. Write a sentence of interpretation for the “slope” coefficient in the model.

```{r}
cats_model2 <-lm(Bwt~Sex, data = cats)
summary(cats_model2)
```

```
Call:
lm(formula = Bwt ~ Sex, data = cats)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.90000 -0.25957 -0.05957  0.30000  1.00000 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.35957    0.06051  38.997  < 2e-16 ***
SexM         0.54043    0.07372   7.331 1.59e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4148 on 142 degrees of freedom
Multiple R-squared:  0.2745,	Adjusted R-squared:  0.2694 
F-statistic: 53.74 on 1 and 142 DF,  p-value: 1.59e-11
```

Question 2

I have a large dataset on house prices in Frederick from 2019. Two of the variables in the dataset are:

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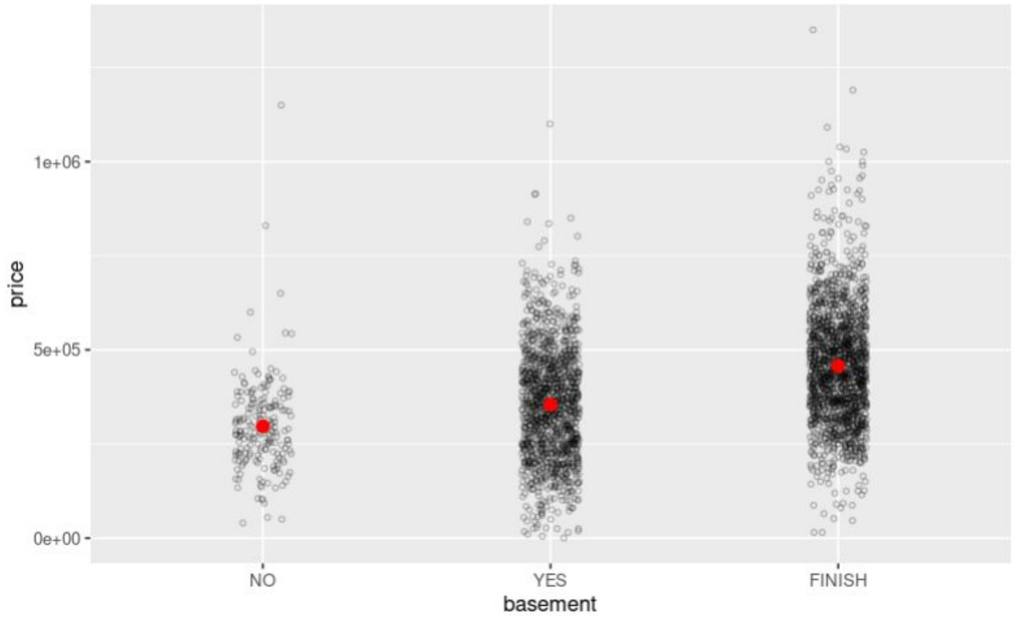
a. Does it look as though having a basement, finished or unfinished, makes a difference to the sales price of a house in Frederick?



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Residuals:
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---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 148300 on 3281 degrees of freedom
Multiple R-squared:  0.1261,	Adjusted R-squared:  0.1256 
F-statistic: 236.8 on 2 and 3281 DF,  p-value: < 2.2e-16
```

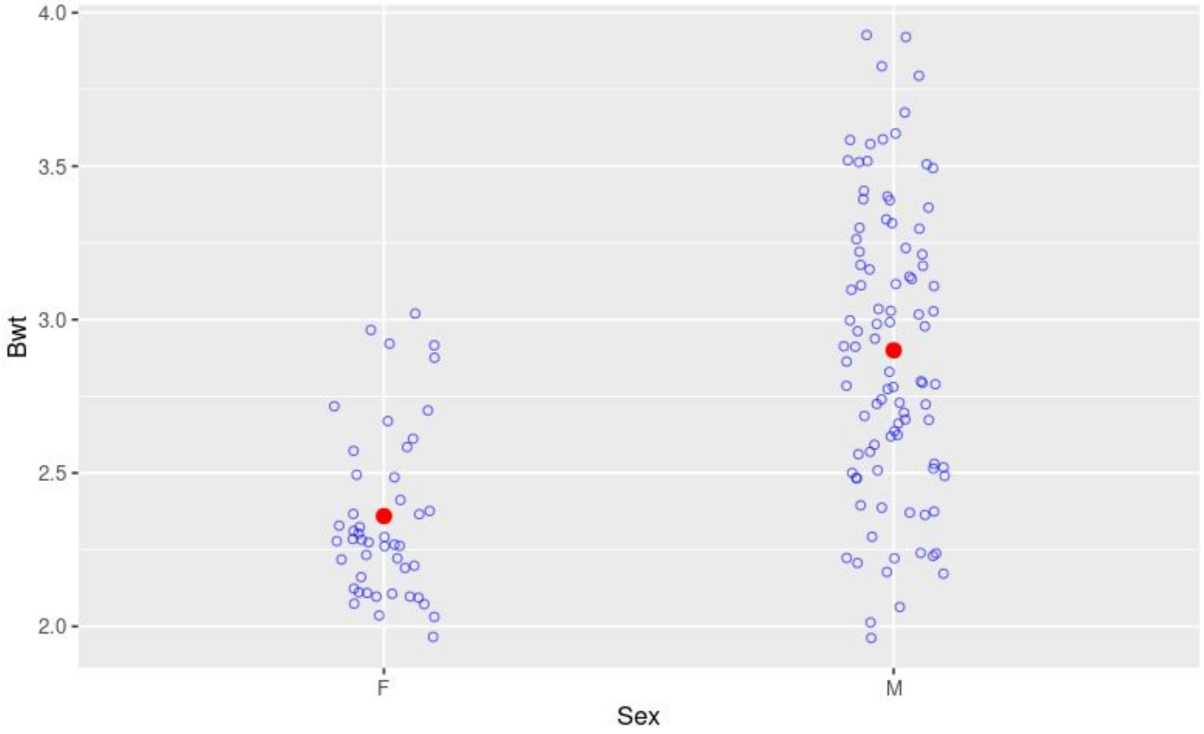
d. Write the regression equation for this model.

e. You should have *three* coefficients in your equation. Write a sentence of interpretation for each one.

Group 10 - Question 1

Researchers collected body measurements on a sample of 144 domestic cats. The cats were all adult, over 2kg in body weight. Variables that were recorded include the weight of each cat's heart (`Hwt`), in grams, the body weight (`Bwt`) of each cat, in kg, and the sex of each cat, (`Sex`).

Here is a graph of the cats' body weights grouped by sex.



a. What am I using as a predictor variable?

a. What am I trying to predict? (What is the response variable?)

Here is some summary output for a regression model I made that matches the situation in the previous graph.

c. Write the linear regression equation.

d. Write a sentence of interpretation for the “intercept” coefficient in the model.

e. Write a sentence of interpretation for the “slope” coefficient in the model.

```{r}
cats_model2 <-lm(Bwt~Sex, data = cats)
summary(cats_model2)
```

```
Call:
lm(formula = Bwt ~ Sex, data = cats)

Residuals:
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Coefficients:
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(Intercept)  2.35957    0.06051  38.997  < 2e-16 ***
SexM         0.54043    0.07372   7.331 1.59e-11 ***
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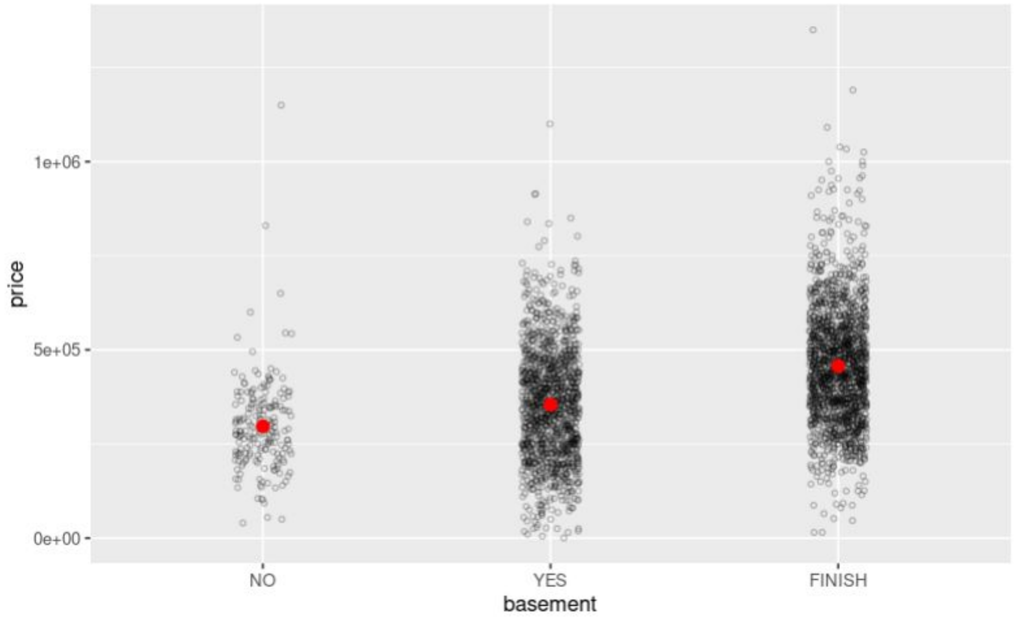

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